

ANAND BHALERAO

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Experience

FIS Global

Pune, India

Software Engineer 2

Jan 2026 - Present

- Designed and shipped **real-time session revocation** across multi-tenant systems serving **40+ banking clients** with zero downtime, containing a **\$2M fraud vector**; built atomic invalidation using Spring + PostgreSQL.
- Architected field-level AES encryption using a custom **MyBatis interceptor** for runtime SQL rewriting across 80% of application queries, **securing PII** across 500K+ records at **0.8ms overhead** paired with **blind indexing**, reducing lookup time from **47s to 60ms** compared to a naive decrypt-and-scan approach.
- Engineered a reusable **zero-downtime AES key rotation framework** for PII-bearing data, featuring parallel chunked migration, resumable execution, encryption versioning, and **idempotent re-encryption**; enabled safe rotation of **500K+ encrypted records** across 10+ deployment environments.

Software Engineer 1

Jul 2024 - Jan 2026

- Led the **end-to-end Zelle SSO integration** for 12+ banking clients, implementing user-state **reconciliation** that synchronized external Zelle records with the bank's source of truth before SSO, alongside automated lifecycle workflows.
- Resolved **burst-traffic row-lock contention** by replacing a single-row sequence table with **block-based cached allocation**, eliminating request timeouts under load.
- Replaced high-overhead in-memory pagination by implementing database-level pagination, sorting, and filtering; reduced data fetched per request from **90K rows to 10 rows**, improving latency for **100K+ datasets**.
- Redesigned multi-entity session handling**, eliminating recurring user lockouts and enabling seamless account switching without redundant re-authentication.
- Integrated Outseer behavioral biometrics** into the authentication layer, enabling risk-based step-up authentication from client-side device signals.

Software Development Engineer Intern

Jan 2024 - Mar 2024

- Developed an end-to-end Electronic File Delivery feature, building **8 Angular screens** and **6 Spring REST APIs** to manage user preferences within a service-oriented architecture.
- Engineered a homepage interstitial for mandatory contact updates, implementing admin-controlled retry logic to ensure high compliance.

Projects

Distributed UPI Payment Simulator

Demo Video

Sarvatra Technologies

- Architected a **distributed UPI payment simulator** modeled on NPCI switch architecture using Spring Boot, Redis, and **RabbitMQ**.
- Implemented secure authentication using JWT for session management and **FIDO2/WebAuthn** for passwordless authentication with public-key cryptography.
- Built fault-tolerant payment processing with **idempotency**, **Saga-based distributed transactions**, and Redis-backed **Bloom filters** reducing database lookups by 92%.
- Instrumented end-to-end observability (OpenTelemetry + Prometheus) and load tested up to **1,800 TPS**, achieving **85ms p95** and **140ms p99** latency under sustained load.

Education

Pune Institute of Computer Technology

Jun 2020 - Jun 2024

B.E in Computer Engineering (CGPA - 8.94)

Pune, Maharashtra

Achievements

- 1100+ problems** solved across LeetCode (peak 1763), CodeChef (AIR 145, 496 in Starters), and Codeforces.
- Finalist in Smart India Hackathon '22.

Technical Skills

Languages: Java, Kotlin, Typescript

Application Development: Spring Boot, MyBatis, Hibernate, JPA, Angular, Compose Multiplatform

Databases & Cache: PostgreSQL, Oracle, MySQL, Redis

Infrastructure & Observability: RabbitMQ, OpenTelemetry, Prometheus